CartONG

Mobiliser les données
Renforcer les organisations
Accompagner le changement

Participatory Mapping and Open Data for Community Resilience

Geo4Impact conference- May 2025

Maeve de France, Head of Programs, CartONG

Table of Contents

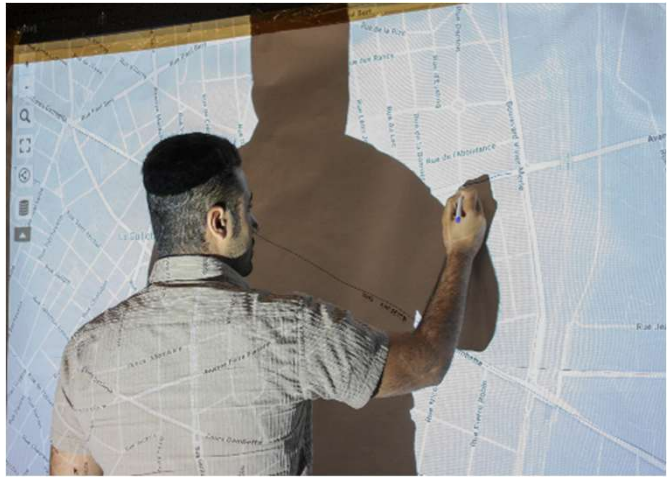
- Who we are
- Why use participatory mapping?
- Practical use cases
- Relevant resources

Who we are

Who we are



Founded in 2006, CartONG is an international NGO based in France, specialized in **mapping and information management for humanitarian aid and international development.**



Our mission : Assist humanitarian actors, development organizations, and public institutions in improving **the quality, accountability and efficiency of their projects, by leveraging geographical and non-geographical data.**

Our services



We equip organizations



We strengthen capacities



We reinforce the sector

Our expertise



Data collection

- Survey-centered tools
- Map-centered tools
- Database development
- Participatory mapping
- Subjective mapping



Web Applications

- Customized geoportals
- Tailored applications
- Seamless integration



Mapping, GIS and data visualization

- Geospatial analysis
- Static maps
- Dynamic maps
- Dashboards
- Remote sensing



Information management strategies

- Assessments of IM processes
- Data literacy
- Responsable data

Why use participatory mapping?

Participatory maps…

- are **great tools for decision making, awareness raising, communication and advocacy**
- are a **visual and accessible medium**, speaking to different cultures & languages



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Our approach on the topic

- **Engage** local communities
- Overcome the **lack of information**
- Build **sustainable** projects

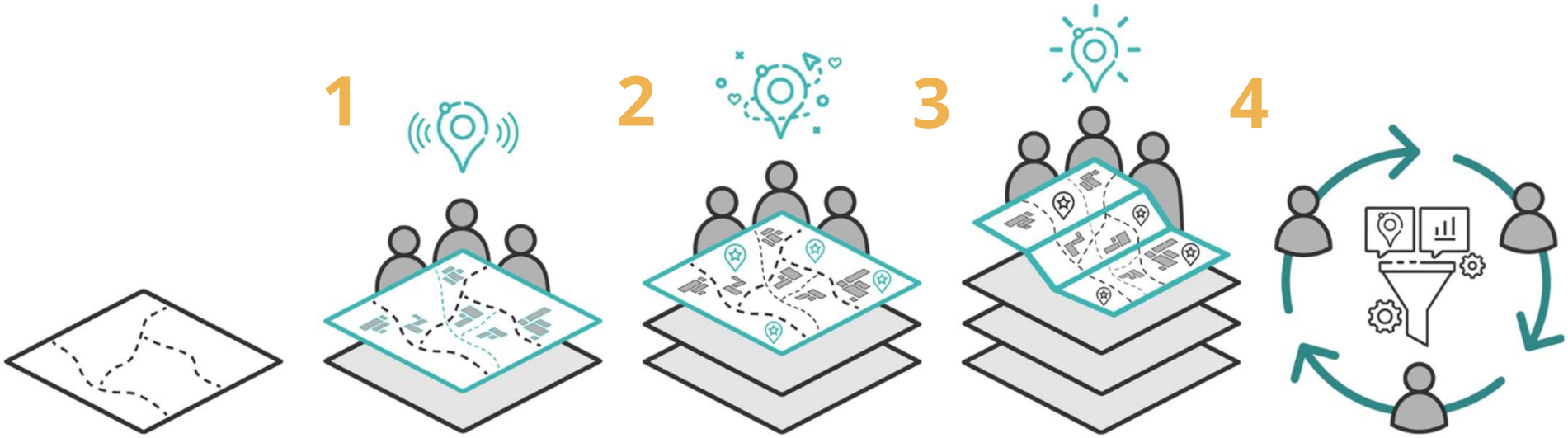
#open data #open source



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4 complementary steps

1. **Mapping** the area of interest
2. **Completing** the map with thematic and/or subjective data
3. **(Co-)producing** mapping products
4. **Ensuring the sustainability** of the developed resources and tools.



Many different tools and methods

Paper participatory mapping



Digital participary mapping



Sensitive mapping



Field Papers



OsmAnd
MAPS & NAVIGATION

OPEN DATA KIT

umap

USHAHIDI
RAISE YOUR VOICE

GOGO carto

3 practical use cases

Participatory Mapping in Mayotte's Shanty Towns



Country: Mayotte (France)

Main partner: Médecins du Monde (MdM)

Objective: Mobilize residents of Longani and Dzoumogné and improve their access to water and the local infrastructure management



Description :

- Participatory methodology, based on simple mapping tools that enabled residents to actively contribute to the production of geographic information through mapping workshops
- CartONG transformed this data into maps, producing 6 maps for each neighborhood

Residents developed micro-projects based on the maps to improve their living conditions . A steering committee of residents and neighborhood associations was formed to prioritise and implement these citizen initiatives.

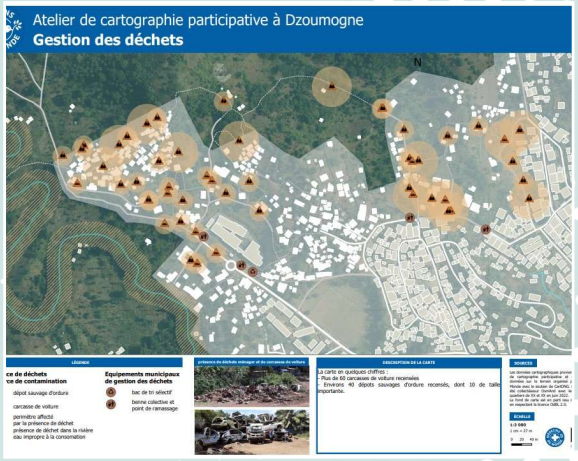
→ Maps as a tool of ownership and transformation.

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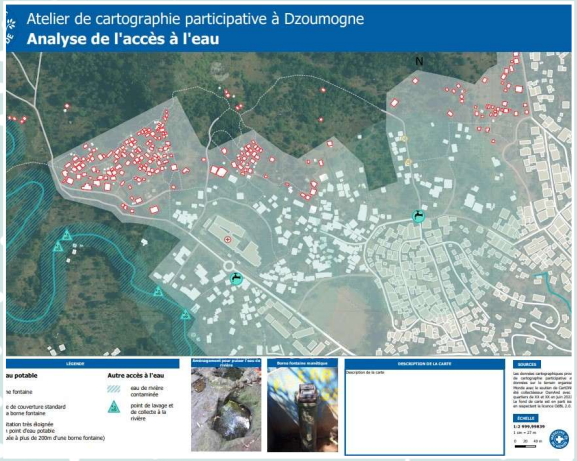
"Everyone is aware of [how the question of water access is a] problem but we know that making it visible on a map would be more concrete and therefore useful to bring about change.

Our approach is to go to the residents to understand their problems [...] we were convinced that a project of this type could motivate the residents, mobilize them to improve their living conditions - it was also an opportunity to get to know each other better and create a link with the residents »

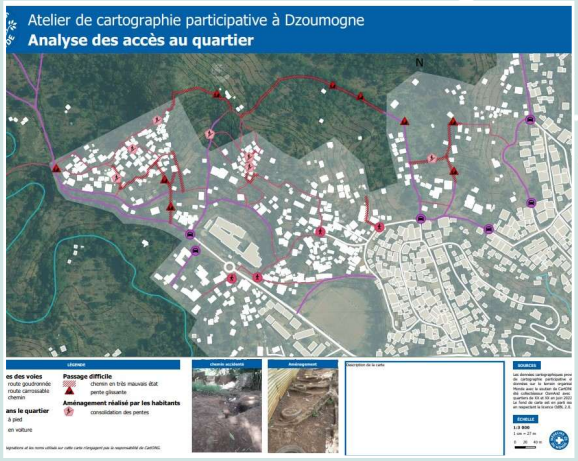
MDM project manager.



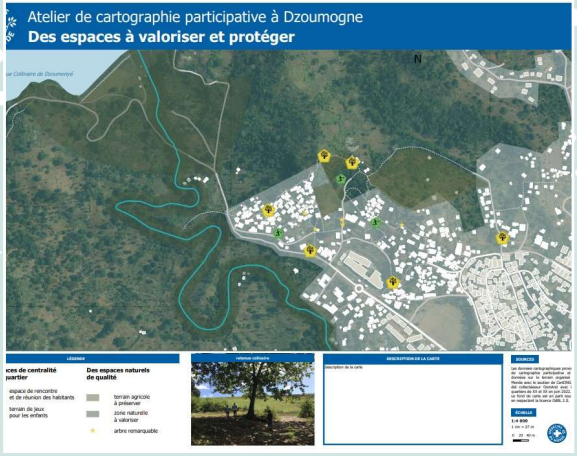
Waste management



Water access analysis



Access to the neighbourhood



Spaces to enhance and protect

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Community-based mapping for climate-resilient development



Country: Tajikistan

Main partners: GIZ/ German cooperation, Little Earth (local partner)

Objective: improve the sustainable and climate-resilient development of local communities



Before and After baseline mapping





Description :

- Raise the capacities of an OpenStreetMap community in Tajikistan, to support community-based mapping activities throughout the country;
- Create detailed geodata for CSOs and for communities themselves of mountainous regions of Tajikistan, of the territories affected / at risk of climate change, through a gender based data collection approach

The project helped map 160 villages, improving the sustainable and climate-resilient development of local communities through better geographical knowledge

→ Maps as a tool of shared understanding and sustainable development



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Community-led anticipatory action in Dar es Salaam



Country: Tanzania

Main partners: H2H network (funder), Open Map Development Tanzania (local partner)

Objective: strengthening community-led anticipatory action mechanisms relating to flood preparedness, for the residents of Mnyamani Ward, Dar es Salaam

Anticipatory action (AA) is acting ahead of predicted hazardous events to prevent or reduce acute humanitarian impacts before they fully unfold

Forecast + Action



Description :

- Engaging with national experts (OMDTZ) and community leaders, to take an active role in disaster preparedness through participatory data collection, and mapping of flood-prone areas
- Building of a dashboard with critical flood risk data, combining community-reported information with predictive analytics and real-time meteorological data

The project piloted an approach of enhanced flood preparedness, streamlining early warning systems and improving coordination among stakeholders

→ Maps an an innovative combination of advanced sources of data & tech, combined with local knowledge

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Interesting resources

Interesting resources

- The CartONG resource portal and its learning corner (https://www.im-portal.org/learning-corner)
- Get in touch if you would like to hear more!



